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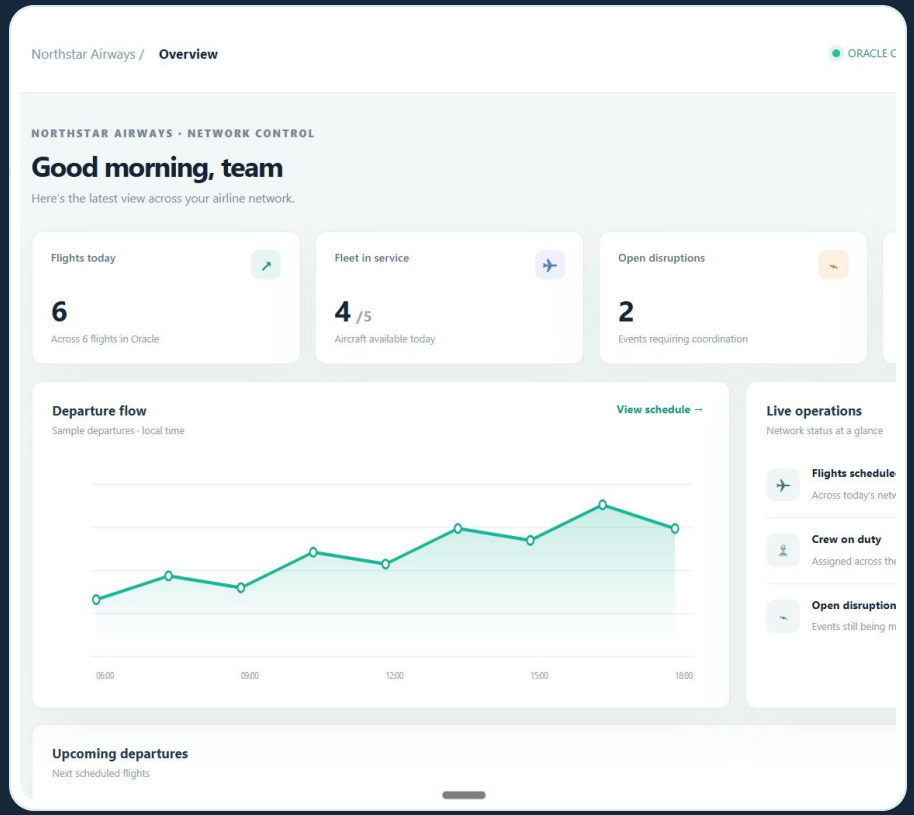
MODERN AIRLINE LOGISTICS & SUSTAINABILITY MANAGEMENT SYSTEM

Database project

Pranali Paygonda Patil

25BCE2957 | B.Tech Computer Science and Engineering

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REVIEW PRESENTATION

Flights • fleet • crew • cargo • sustainability

Project overview

Abstract

A relational database and browser console that connect airline operations records around the flight, making schedules, resources, cargo conditions, disruptions and emissions easier to review together.

WHY THIS PROJECT

One linked view

Flights connect to aircraft, crew assignments, cargo, passengers, disruption events and fuel / CO_2 logs.

Better data quality

Primary and foreign keys, required values and checks help prevent broken links and invalid records.

Sustainability visibility

Flight-level fuel and CO_2 fields create a starting point for operational comparison and reporting.

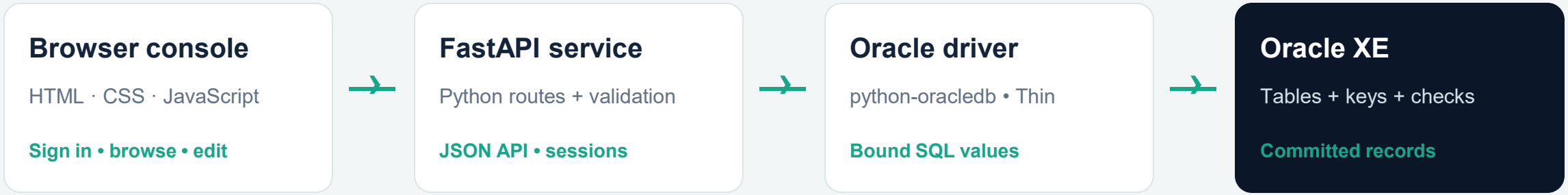
9 **relational entities**

AIRCRAFT • FLIGHT • CREW • FLIGHT_CREW • PASSENGER • LOYALTY_TIER • SPEC_CARGO • DISRUPTION • EMISSIONS

Current build: FastAPI + python-oracledb (Thin mode) + Oracle XE.

How the system works

REQUEST PATH



DATA LIFECYCLE

1 Sign in

Local account protects operational routes

2 Read

Console requests records from Oracle

3 Change

Create, update or delete through mapped API routes

4 Refresh

Changes commit; views refresh from database

Operational data routes require sign-in; this coursework prototype runs locally at 127.0.0.1:8001.

Implementation and practical value

BUILT AND DEMONSTRATED

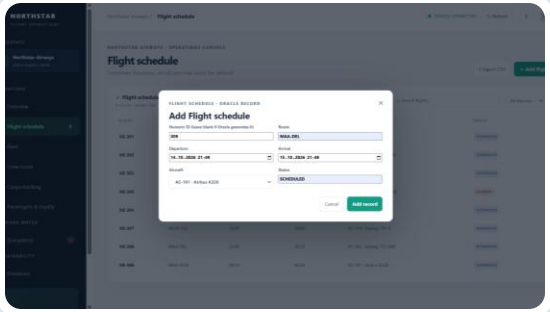
- **Flight CRUD**
  A flight was added and updated in Oracle. The delete screenshot shows the confirmation prompt, not the completed deletion.
- **Operations modules**
  Overview, flight schedule, fleet, crew, cargo, passengers, disruptions and emissions views.
- **Local implementation**
  FastAPI serves the browser console; python-oracledb connects the service to Oracle XE.

SUSTAINABILITY INSPIRATION

Make impact visible where operations are recorded: fuel and CO_2 per flight, plus temperature requirements and readings for sensitive cargo.

BUSINESS USE

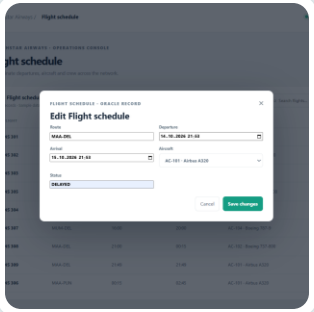
Shared flight-linked records can reduce lookup across separate lists, help cargo teams spot temperature exceptions, and give analysts a base for route and fleet comparisons.



ADD RECORD



ORACLE ROWS



UPDATE RECORD

Potential value is a project hypothesis, not a measured financial or emissions outcome. Demo data is illustrative.

Future implementation roadmap

The next steps move from a safer prototype to monitored operations and validated reporting.

NEAR TERM

Harden the prototype

- Role-based access and audit trail
- Stronger validation + explicit ID strategy
- Verify delete and access-control tests
- Backups, monitoring and error handling

OPERATIONS

Connect live workflows

- Crew assignment with duty/rest checks
- Cargo sensor feeds + threshold alerts
- Disruption history and response tracking
- Pilot with operations and cargo users

SCALE + REPORT

Prove business outcomes

- Validated emissions factors and boundaries
- Route/load-normalized fuel reporting
- Approved airport and maintenance links
- Cloud deployment and recovery plan

Before claiming impact, establish a baseline and validate data completeness, emission factors, fuel units and cargo temperature records.

Project demo video

GOOGLE DRIVE LINK

https://drive.google.com/drive/folders/1iJCeO4C2KEm1-Avz9S9sdOv1OaXjlvBR?usp=sharing

SUGGESTED DEMO FLOW

1 Sign in → 2 Open flight schedule → 3 Add or update a flight → 4 Show Oracle record → 5 Visit cargo + emissions

The demo URL was not present in the supplied project files, so add it before submission.

THANK YOU

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Pranali Paygonda Patil • 25BCE2957